SMART INDIA HACKATHON 2026

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TITLE PAGE

- **Problem Statement ID -** SIH26205
- **Problem Statement Title -** Student Innovation- Ideas to address the growing pressures on the city’s resources, transport networks, and logistic infrastructure.
- **Theme -** Transportation and Logistics
- **PS Category -** Software
- **Team ID -** 191988
- **Team Name -** Syntax Error



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SPACE2PARK

EVENT-AWARE SMART PARKING

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Proposed Solution:

SPACE2PARK: Intelligent AI-Powered Event-Based Smart Parking Network.

Large events create temporary parking demand that existing nearby parking capacity may not satisfy. Meanwhile, some private and commercial spaces may remain unused. **SPACE2PARK** connects event attendees with verified, temporarily available parking spaces. It estimates demand, matches vehicles to suitable spaces, manages reservations and tracks occupancy through digital check-in/check-out.

USP: Predict event-specific parking demand and dynamically activate + allocate nearby distributed spaces before congestion occurs.



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TECHNICAL APPROACH

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System Architecture

Frontend (Client Side)
User Interface and Experience

- **React 18+** — Component-based UI development
- **Vite** — Fast build tool and dev server
- **JavaScript (ES6+)** — Modern JavaScript features
- **Tailwind CSS** — Responsive and modern UI
- **React Router DOM** — Client-side routing
- **Context API** — Global state management
- **Framer Motion** — Smooth animations
- **Lucide React** — Icon library

Backend (Server Side)
Business Logic, APIs and Real-time Services

- **Node.js** — Runtime environment (Scalable & Non-blocking)
- **Express.js** — REST API framework
- **JWT** — Authentication & Authorization (RBAC)
- **REST APIs** — CRUD operations and business logic
- **Multer** — File upload handling (images & documents)
- **Socket.io** — Real-time updates (slot status, occupancy, notifications)

API Requests (JSON)
HTTPS / REST / WebSocket

Database & Storage
Data Storage and Management

PostgreSQL (Primary Database)
- Transactional booking data
- ACID properties
- Geospatial data (PostGIS)
- High concurrency support

Redis
(Caching & Locks)
- Session management
- Real-time data cache
- Atomic booking locks
- TTL for temporary holds
- Pub/Sub for live updates

Cloud Storage
(Images & Documents)
- Parking space images
- KYC documents
- Event flyers
- Scalable storage (AWS / GCP / Azure)

Response Data (JSON)
API Response / Real-time Updates

End-to-End Workflow for All Stakeholders

Start → Open Platform (App / Website) → Select Role

Event Host (Organizes Event)
- Sign Up / Login (Institute verification)
- Create Event Listing (event venue, dates, capacity, pricing, requirements)
- AI Demand Prediction (estimate parking demand)
- Submit Event (for verification)
- Track Status in Dashboard (Pending / Approved / Rejected)
- Event Approved? — No → Edit & Resubmit; Yes → Event Goes Live (Real-time updates)
- Receive Confirmation (event live + analytics dashboard)
- Monitor Event (view bookings, footfall, reports)

Admin Team (Verifies Content)
- Receive Event Submission
- Revive Event Details (rules, authenticity, venue, documents, pricing configuration)
- Valid Event? — No → Reject Event (with reason); Yes → Approve Event
- Event Goes Live (on platform)
- AI Insights for Admin (predicted demand, risk alerts, capacity recommendations)
- Trigger Notifications (to event hosts and relevant teams)
- Event Visible to Parking Seekers

Space Owner / Vendor (Parking Owners / Businesses)
- Sign Up / Login (as homeowner / business)
- Receive Event Notification (new event near you within 2 km)
- List Parking Space? — No → Ignore; Yes
- Add Parking Space Details (address, type, capacity, price, photos, availability time slots)
- AI Suggested Pricing (based on demand, location, event type)
- Submit for Quick Verification (automatic + admin check)
- Listing Approved? — No → Update Details & Resubmit; Yes
- Listing Goes Live (visible to parking seekers)
- Manage Bookings (view upcoming, current, past)
- Receive Payments (after event completion)
- View Earnings & Reviews (ratings from parkers)

Parking Seeker (Finds & Books Parking)
- Sign Up / Login (as vehicle owner)
- Search for Event (browse upcoming events or search by name)
- View Event Details (date, venue, expected crowd)
- AI Parking Recommendations (rank spaces by distance, price, availability, and facility)
- View Available Parking Spaces (on map with real-time availability)
- Apply Filters (price, distance, type, etc.)
- Select a Parking Space (view details, photos, reviews)

Real-Time Booking & Concurrency Manager
- Check Real-Time Availability (atomic check with database)
- Space Available? — No → Show updated availability (refresh list); Yes
- Temporarily Lock Space (hold for 5 minutes)
- Make Payment (UPI / Card / Wallet)
- Payment Successful? — No → Release Lock (automatically); Yes
- Confirm Booking (update occupancy in real-time)
- Receive Booking Confirmation (booking ID, QR code, directions)
- Navigate to Parking Location (using in-app maps)
- Arrive & Park (show QR code / verification)
- Exit Parking (automatic / manual check-out)
- Rate & Review (share your parking experience)

AI & Core Logic Modules

Intelligent Demand Prediction & Smart Allocation

1. Event Demand Prediction (AI/ML)

Input Features
- Expected attendance
- Event type/category
- Date & time
- Historical booking data
- Location & distance
- Current occupancy
- Nearby space availability
- Weather & road conditions

Model
Time Series Regression (LSTM)
Trained on historical event and parking data

Output
- Predicted parking demand
- Peak time prediction
- Recommended capacity
- Demand heatmap
- Confidence score

2. Smart Allocation Engine

Input
- Predicted demand D
- Available spaces S
- Distance to event d_i
- Price p_i
- Real-time availability a_i
- Space type & capacity c_i
- Historical preference h_i

Allocation Algorithm

$$\text{Score}(i) = w_d \cdot \frac{1}{1+d_i} + w_p \cdot \frac{1}{p_i} + w_a \cdot a_i + w_c \cdot \frac{c_i}{c_{max}} + w_h \cdot h_i$$

Assign top $-k$ spaces with highest $Score(i)$

Subject to: $\sum_{i=1}^{N} x_i \le D, \quad capacity_i \ge required_slots$

$x_i \in \{0, 1\}$

Output
- Ranked parking options
- Optimal allocation
- Dynamic pricing
- Suggested spaces
- Load balancing across multiple locations

3. Booking & Concurrency Management

Atomic Booking Flow
1. Check availability (atomic query)
2. Temporarily lock space (Redis + DB)
3. Process payment (UPI / Card / Wallet)
4. Confirm booking (transaction commit)
5. Release lock on failure/timeout

State Transitions (Slot Lifecycle)

Available → Locked (5 min TTL) → Booked

Locked → Payment Failed / Timeout (Release Lock) → Automatically Release Lock (After event/end time)

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FEASIBILITY AND VIABILITY

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FEASIBILITY
(Technological & Operational)

- Uses widely available technologies (web/app, GPS, maps, QR, digital payments)
- Leverages existing infrastructure and open APIs
- MVP supports event creation, space listing, demand prediction, booking & QR verification
- Scalable architecture for multiple events and cities
- Deployment-ready integrations can be added incrementally

KEY CHALLENGES
(Risks)

- Verification of genuine parking spaces and owner trust
- Dynamic and uncertain demand during events
- Concurrent bookings / stale availability
- Integration with entry/exit systems (QR; ANPR/FASTag as future integration)
- Regulatory and local permissions
- No-shows and last-minute cancellations

STRATEGIES
(Mitigation)

- Owner KYC, geo-verification and admin approval
- AI demand prediction + supply-demand allocation
- Atomic reservation + temporary lock to prevent double booking
- Start with QR-based access; future ANPR/FASTag integration
- Collaborate with local authorities and event organizers
- Cancellation policies + real-time slot management

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IMPACT AND BENEFITS

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ASPECT	IMPACT ON TARGET AUDIENCE	KEY BENEFITS / MEASURABLE KPIs
Event Attendees (Users)	Convenient, reliable pre-booked parking during events.	• Parking search time ↓ • Pre-booking completion rate ↑ • Verified-space availability ↑ • Booking success rate
Parking Space Owners (Homes, Shops, Offices)	Monetize unused spaces during event demand peaks.	• Space utilization ↑ • Owner earnings per event • Listing fill rate • Verified active spaces
Event Organizers	Better parking allocation and crowd-flow management.	• Event parking demand fulfilled % • Average allocation distance • Real-time availability coverage • No-show / cancellation rate
Local Authorities & City Administration	Data-driven parking and smoother event traffic.	• Roadside/illegal parking incidents ↓ • Average cruising distance ↓ • Parking utilization data • Event traffic-flow indicators
Environment	Reduce unnecessary cruising for parking and improve use of existing spaces.	• Average search distance ↓ • Fuel consumption proxy ↓ • Estimated CO_2 emissions ↓ • Existing-space utilization ↑
Society / Community	More organized, safer and accessible event areas.	• Road-blockage incidents ↓ • Safety/complaint rate • Accessible-space availability • Community participation

Impact is measured through platform KPIs; final city-level reductions require pilot data.

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RESEARCH AND REFERENCES

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RESEARCH FINDINGS

Smart Parking Systems: Real-time availability, occupancy monitoring, navigation and digital payments improve parking management.
Demand Prediction: ML/regression approaches can forecast parking demand using historical and contextual data.
IoT & Automated Parking: Sensors, IoT and computer vision improve occupancy detection but may require dedicated infrastructure.
Urban Parking Research: Key challenges include parking search time, congestion, availability reliability, security and interoperability.

RESEARCH

01. *Smart Parking Systems: Comprehensive Review Based on Various Aspects*
Fahim, A., Hasan, M. & Chowdhury, M.A. | *Heliyon* | **2021**

02. *Smart Parking in IoT-Enabled Cities: A Survey*
Al-Turjman, F. & Malekloo, A. | *Sustainable Cities and Society* | **2019**

03. *Parking Occupancy Prediction and Analysis – A Comprehensive Study*
Channamallu, S.S. & Kermanshachi | *Transportation Research Procedia* | **2023**

04. *Advanced IoT-Integrated Parking Systems with Automated License Plate Recognition and Payment Management*
Pradhan, G., Prusty & Chinara, S. | *Scientific Reports* | **2025**

05. *A Survey of Smart Parking Solutions*
Lin, T., Rivano, H. & Le Mouël, F. | *IEEE Transactions on Intelligent Transportation Systems* | *2017*

Technology & AI Tools: OpenStreetMap, Google Platforms

OUR CONTRIBUTION - SPACE2PARK

EVENT DATA
Attendance • Location • Date/Time • Event Type
↓
AI DEMAND PREDICTION
Predict parking demand & peak requirement
↓
DISTRIBUTED SUPPLY
Discover & verify nearby private/commercial spaces
↓
SMART ALLOCATION
Distance • Price • Capacity • Availability • Demand
↓
RELIABLE BOOKING
Atomic reservation • Temporary lock • Payment • QR verification

Most existing systems optimize existing parking supply. They do not primarily address temporary demand spikes during large events by coordinating distributed, privately available parking spaces.